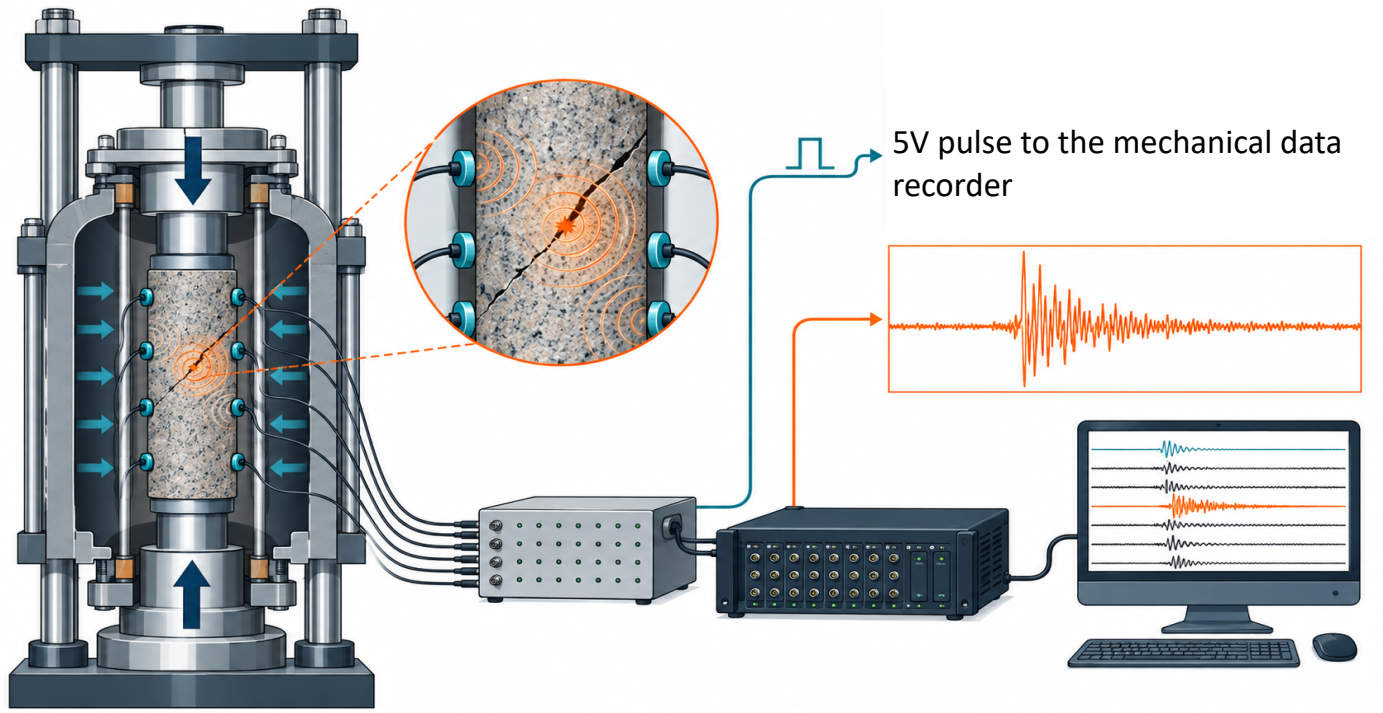
The UCS + AE experiment: what will we record?



Synchronized observations

Mechanical: load, shortening, time.

Acoustic: individual sensor traces, sampling rate, acquisition time.

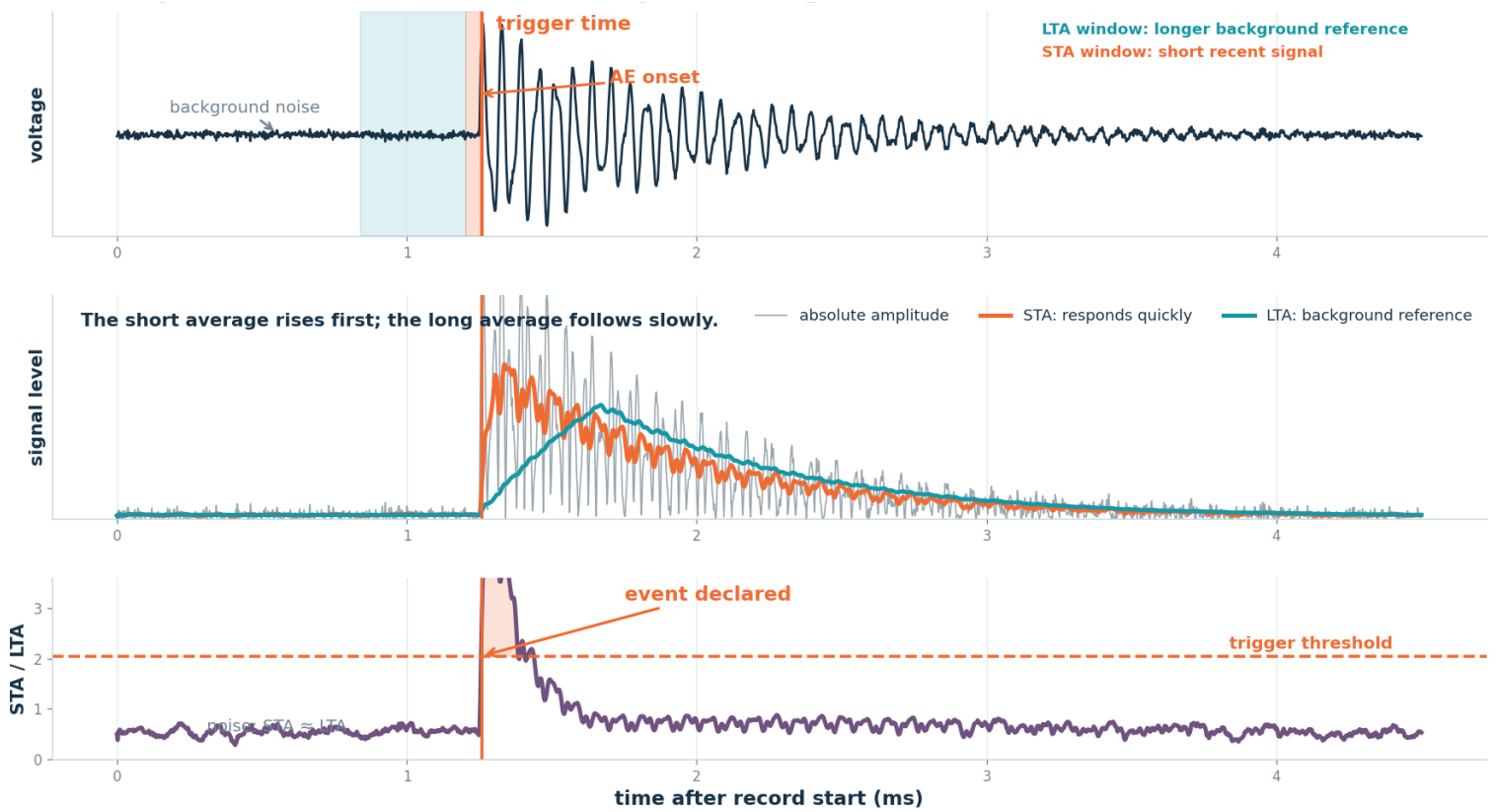
Goal: relate the onset of AE activity to changes in mechanical loading.

Record channel names, units, threshold choices and the mechanical–AE time offset.

DETECTION

The threshold edits the history you think you observed

Activity 1



Low threshold

More small hits — but also more electronic noise and false triggers.

High threshold

Cleaner catalogue — but early and distant microcracks may disappear.

An increase in event rate can be physical, instrumental, or both.

Prompt: which threshold makes the ‘onset of damage’ later?

Schematic signals (editable): same physical sequence, different catalogue completeness.

A catalogue is a compressed, model-dependent representation of damage

WHEN

origin time
t_0

WHERE

x, y, z
+ uncertainty

HOW LARGE

amplitude
energy /
magnitude

HOW

polarity
radiation pattern

What compression removes

Full waveforms contain frequency content, overlapping sources and path effects. A catalogue keeps selected attributes after detection, picking, association and inversion.

Treat the catalogue as a measurement product — not ground truth.

A catalogue is a compressed, model-dependent representation of damage

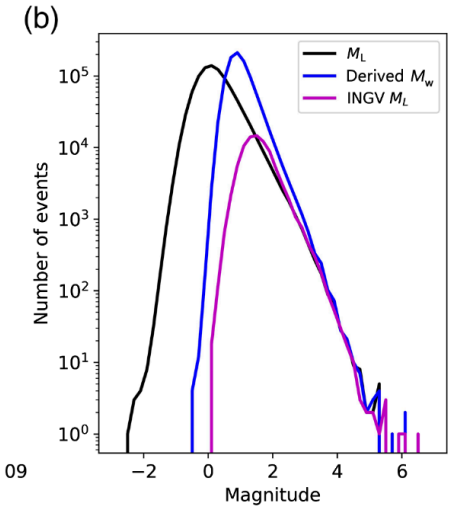
The Seismic Record
A journal of the Seismological Society of America

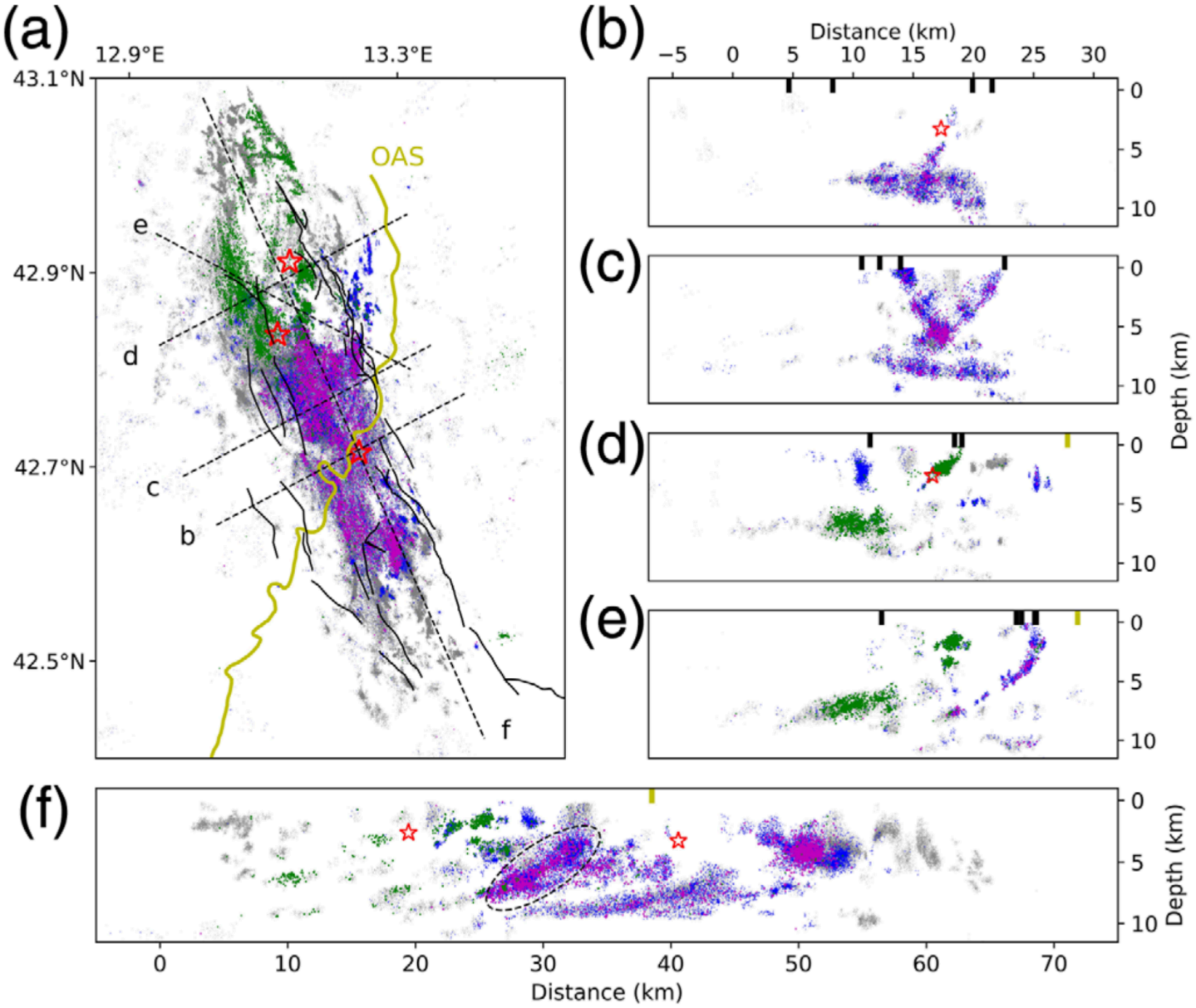
Machine-Learning-Based High-Resolution Earthquake Catalog Reveals How Complex Fault Structures Were Activated during the 2016–2017 Central Italy Sequence

Yen Joe Tan*[1], Felix Waldhauser[2], William L. Ellsworth[3], Miao Zhang[4], Weiqiang Zhu[3], Maddalena Michele[5], Lauro Chiaraluce[5], Gregory C. Beroza[3], and Margarita Segou[6]

The routine catalog by the INGV contains ~**82,000** events.

deep-neural-network-based picker catalog of ~**900,000** earthquakes.





A catalogue is a compressed, model-dependent representation of damage

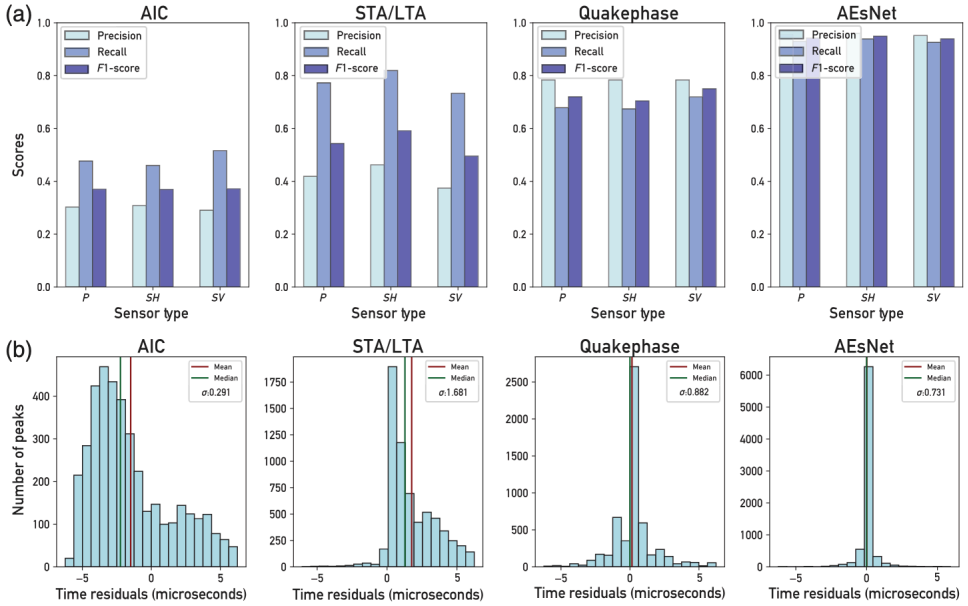
RESEARCH ARTICLE | AUGUST 13, 2025

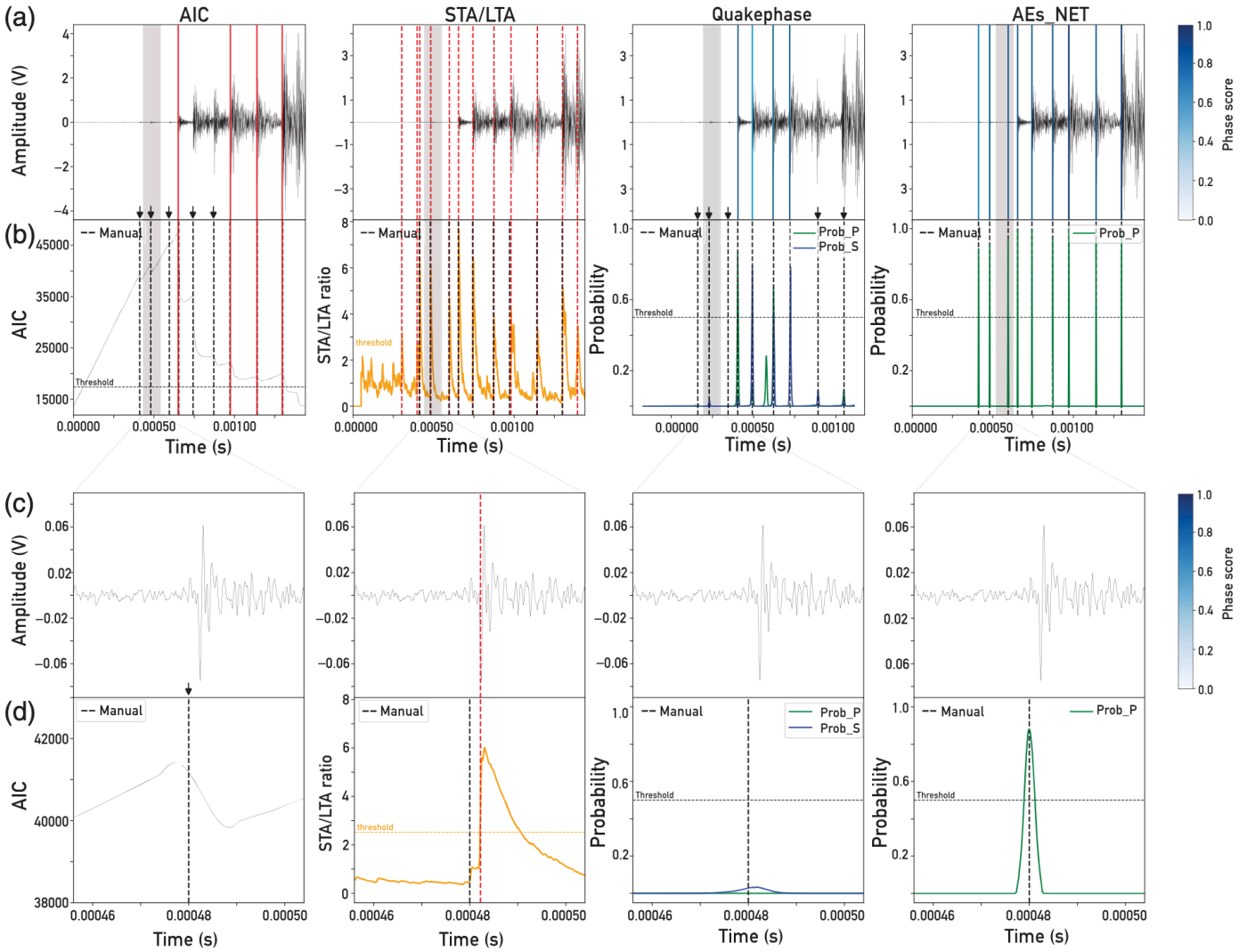
Readapting PhaseNet to Laboratory Earthquakes: AEsNet, a Robust Acoustic Emission Picker Illuminating Seismic Signatures of Different Fault Gouge Materials

Giacomo Mastella; Giulio Poggiali; Federico Pignalberi; Marco Scuderi

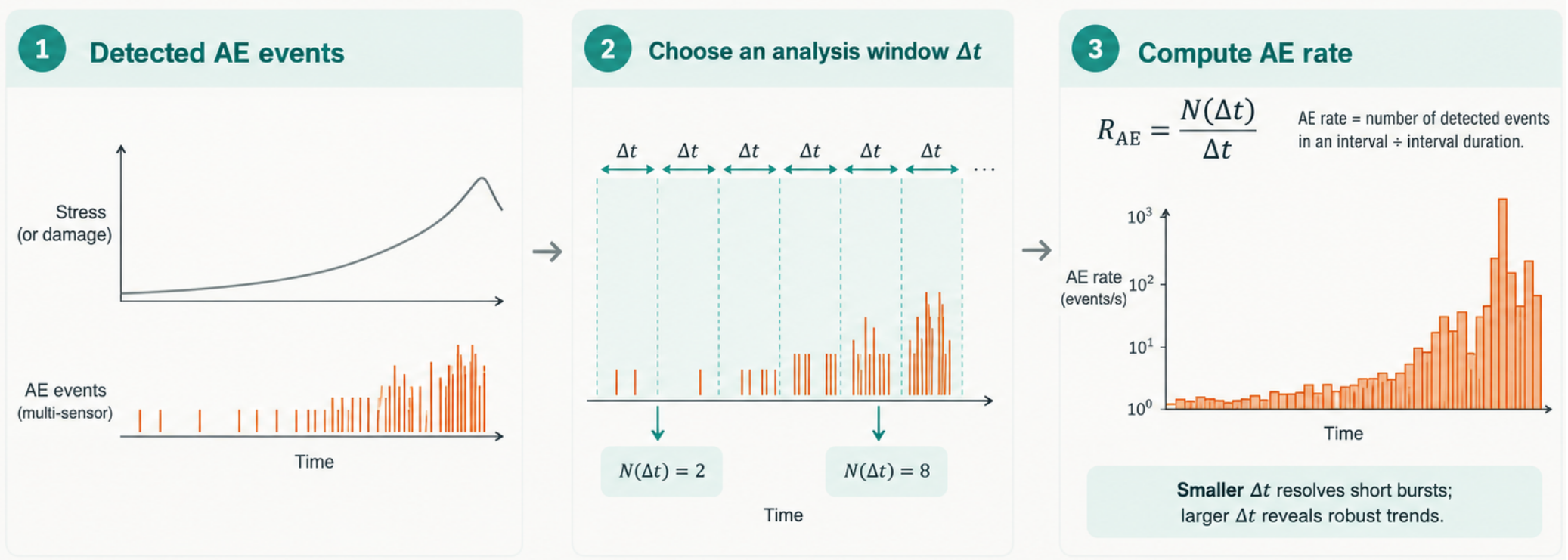
+ Author and Article Information

Seismological Research Letters (2026) 97 (1): 141–160.
https://doi.org/10.1785/0220250025 | Article history





AE rate: the first quantity you will compute in the lab



AE rate = number of detected events in an interval ÷ interval duration.

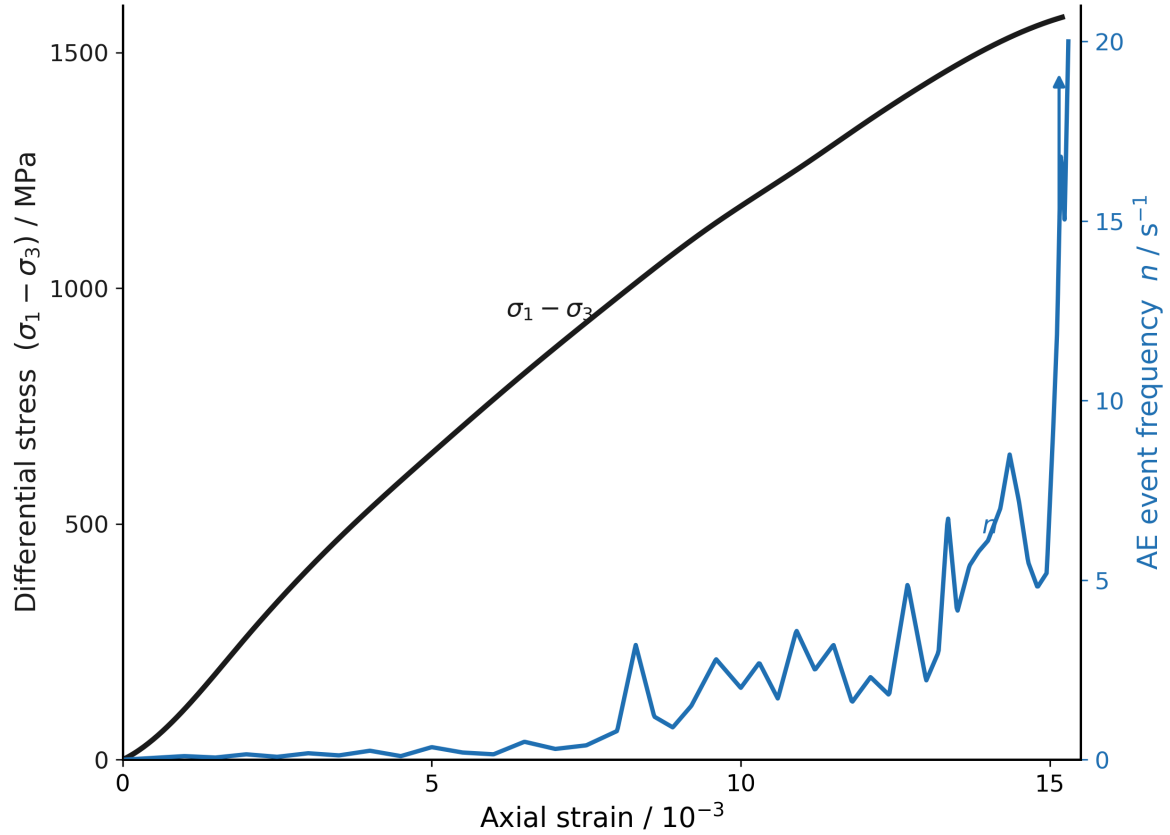
$R_{ae} = N(\Delta t) / \Delta t$

Change Δt and compare the short-term peaks with the robust overall trends.

Use a synchronized time base and state whether you count hits or multi-sensor events.

ACTIVITY

AE commonly accelerates as brittle failure approaches



Redrawn and simplified after Paterson & Wong (2005), Fig. 5.38.

Observation

Event rate rises strongly during dilatant cracking and near peak stress.

But not a clock

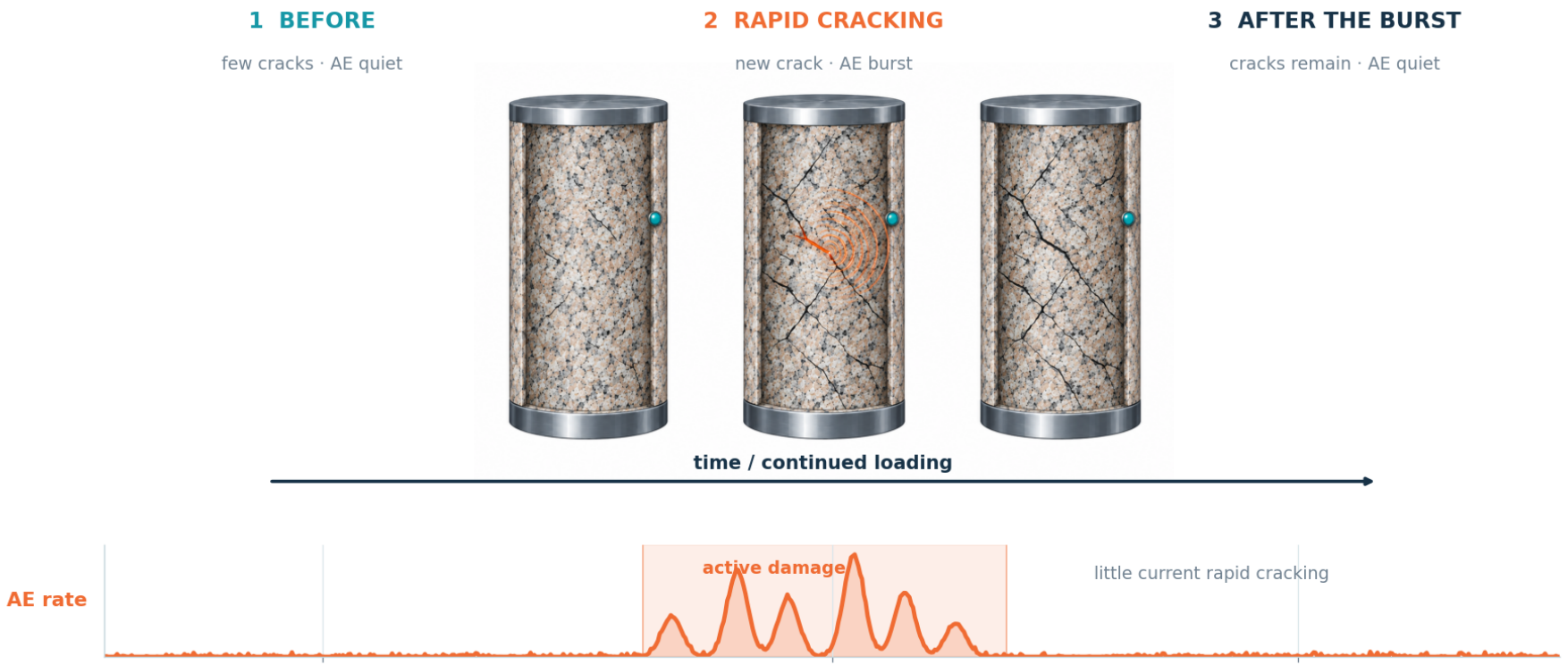
The onset and slope depend on threshold, stress path, confinement, material and catalogue completeness.

rate answers ‘how often?’ — not yet ‘where?’ or ‘what mechanism?’

Paterson & Wong (2005), Fig. 5.38.

ACTIVITY

AE rate is not the same quantity as total damage



DAMAGE PROCESS

- slow crack extension
- rapid microfracture
- grain crushing
- frictional sliding
- pore collapse

AE CATALOGUE

detected rapid events

above threshold
within bandwidth
not hidden by dead time

How do we infer total damage?

Interpretive boundary

A quiet interval can mean little damage, slow damage, weak coupling, attenuation, or an incomplete catalogue.

AE rate is not the same quantity as total damage

1 BEFORE

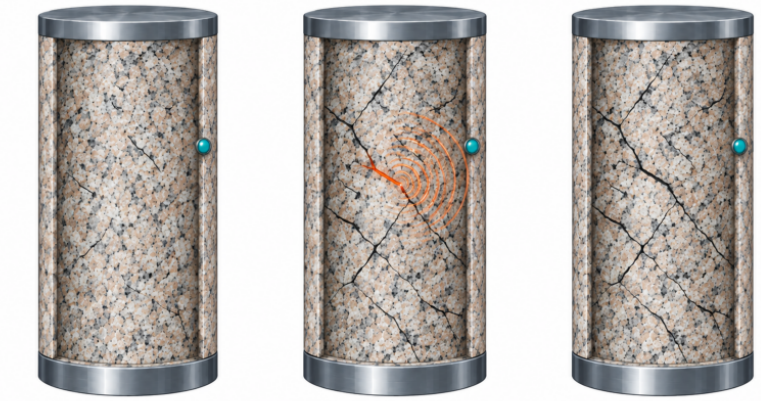
few cracks · AE quiet

2 RAPID CRACKING

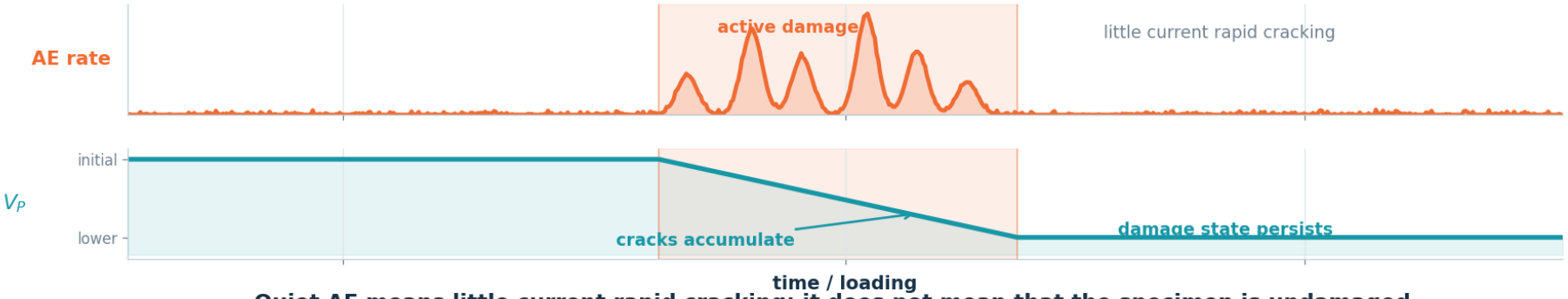
new crack · AE burst

3 AFTER THE BURST

cracks remain · AE quiet



time / continued loading



Quiet AE means little current rapid cracking; it does not mean that the specimen is undamaged.

DAMAGE PROCESS

- slow crack extension
- rapid microfracture
- grain crushing
- frictional sliding
- pore collapse

AE CATALOGUE

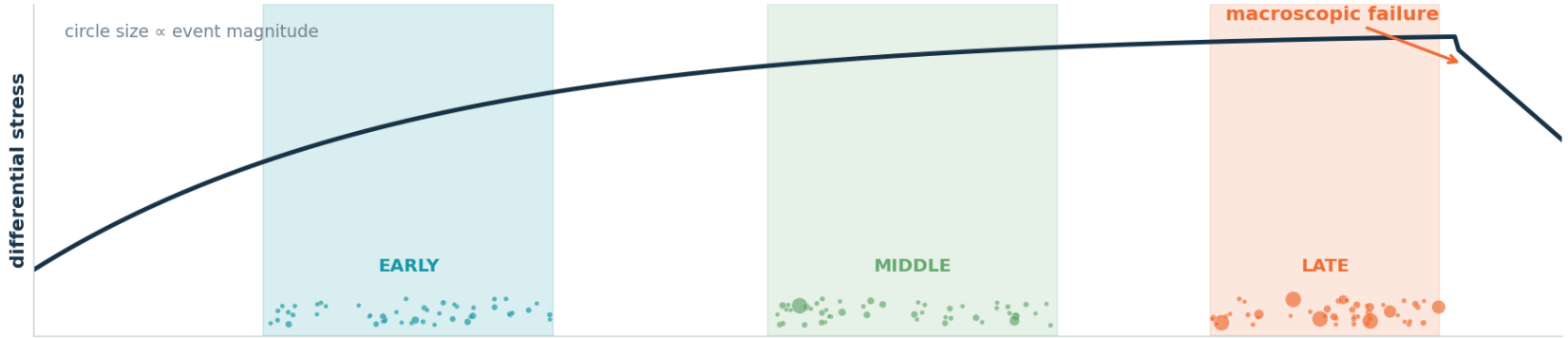
detected rapid events

above threshold
within bandwidth
not hidden by dead time

Many small events, few large events —> b-value

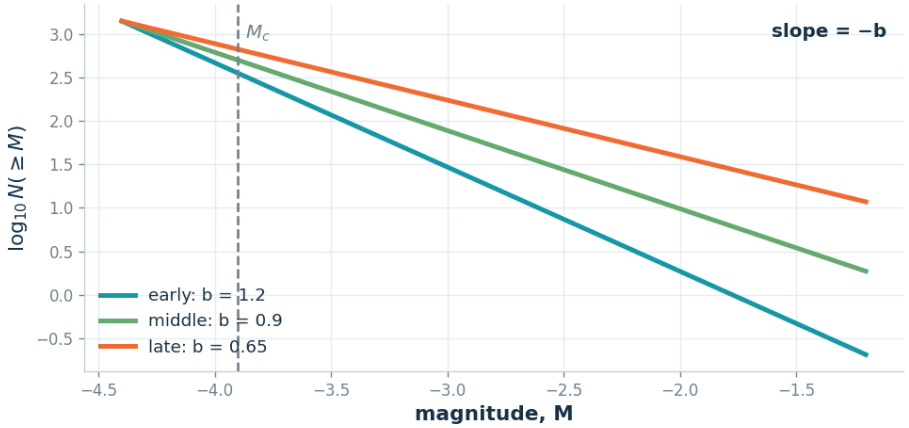
Activity 2

Within one loading cycle, a lower b-value means relatively more large events



$$\log_{10} N(\geq M) = a - bM$$

Magnitude-frequency distribution



What changes in the catalogue?

HIGH b

small events dominate

LOW b

large events become relatively more common

Interpret only above M_c and with a stable detection procedure.

b-value

Controls the relative proportion of small and large events in the observed catalogue.

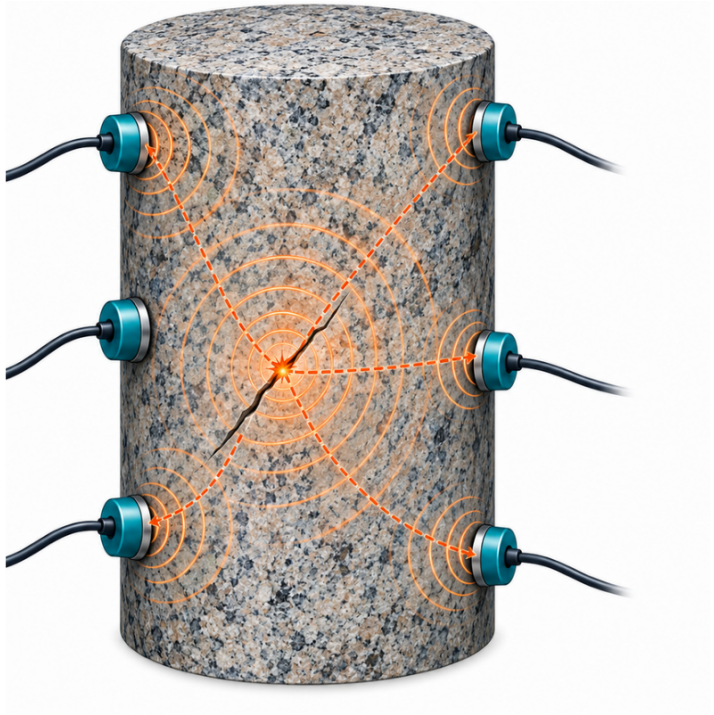
Caution

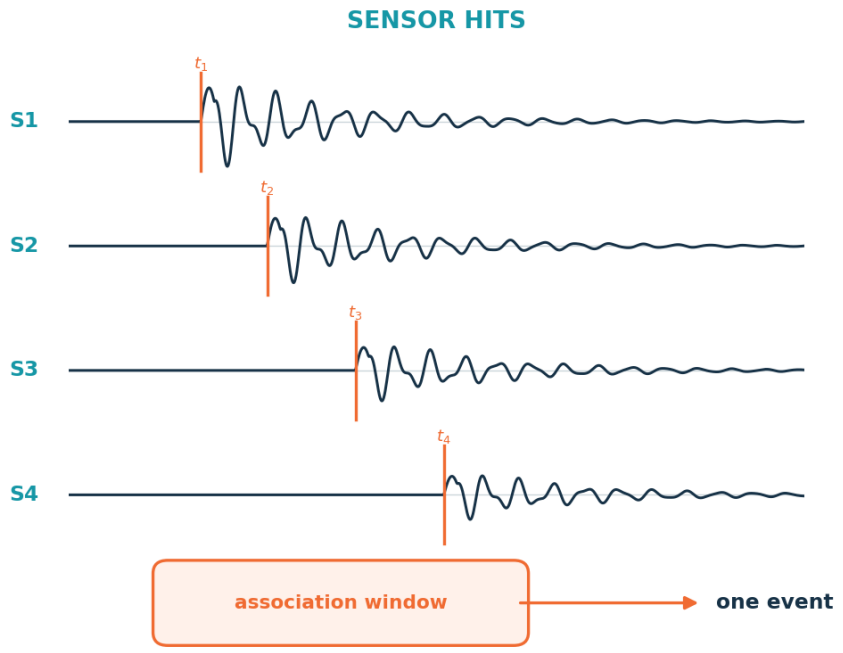
Changing completeness, saturation and time window can imitate physical evolution in b.

Statistical change is evidence to test — not a deterministic failure alarm.

Thompson et al. (2006), Fig. 9; Gutenberg–Richter form.

One crack can produce signals on many sensors





Different path lengths produce different arrival times; the association window groups the hits.

Vocabulary for the lab

Hit: a threshold crossing / recording on one channel.

Event: a physical source inferred by associating hits across sensors.

One event may generate multiple sensor hits.

Never interpret a raw hit count as the number of newly formed cracks.